# *Haifei Zhang*

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*Websites:* ORCID Google Scholar GitHub


## *Profile*

I am currently an Assistant Professor (Maître de Conférences in french) at Jean Monnet University in Saint-Étienne, France. I conduct my research in the Data Intelligence team of the Hubert Curien Laboratory (UMR-CNRS 5516), a joint research unit of Jean Monnet University, CNRS, and the Institution of Optique Graduate School. I teach at Télécom Saint-Étienne, a highly reputed engineering school.

My research primarily focuses on complex data analysis, graph representation learning, and trustworthy AI, with a particular interest in explainable AI (XAI), cautious classifiers, and uncertainty management in machine learning. I am dedicated to developing robust and interpretable AI models that enhance decision-making in high-stakes applications.

## *Education*

**University of Technology of Compiègne** — November 2020 - November 2023
PhD in Computer Science — France
Dissertation: Explainable Cautious Classifiers
Supervisors: Benjamin Quost, Marie-Hélène Masson

**University of Technology of Compiègne** — September 2019 - September 2020
Master's degree in Complex Systems — France

**University of Technology of Compiègne** — September 2017 - September 2020
Engineer's degree in Data Mining — France

**Shanghai University** — August 2014 - July 2017
Bachelor's degree in Information Engineering — China

## *Employment*

**Jean Monnet University** — September 2024 - now
Assistant Professor — France

**University of Technology of Compiègne** — November 2023 - August 2024
Teaching and Research Assistant — France

## *Teaching*

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| **Jean Monnet University** | |
| **Deep learning** | Spring 2025 |
| **Design patterns** | Spring 2025 |
| **Introduction to Git** | Spring 2025 |
| **Web development with PHP** | Fall 2024 |
| **Advanced algorithms** | Fall 2024 |
| **University of Technology of Compiègne** | |
| **Linear algebra basics** | Fall 2023, Spring 2024 |
| **Data analysis and data Mining** | Spring 2022, Spring 2023, Spring 2024 |
| **Statistical methods for engineers** | Fall 2021, Fall 2022, Fall 2023, Spring 2024 |

## *Publications*

### Journal papers

1. Nguyen, Vu-Linh, **Haifei Zhang**, and Sébastien Destercke. "Credal ensembling in multi-class classification." *Machine Learning* 114.1 (2025): 1-62.
2. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. "Cautious classifier ensembles for set-valued decision-making." *International Journal of Approximate Reasoning* 177 (2025): 109328.
3. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. "Cautious weighted random forests." *Expert Systems with Applications* 213 (2023): 118883.

### International conference papers

1. **Zhang, Haifei**. "SHADED: Shapley Value-Based Deceptive Evidence Detection in Belief Functions." *International Conference on Belief Functions*. Cham: Springer Nature Switzerland, 2024.
2. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. "Cautious Decision-Making for Tree Ensembles." *European Conference on Symbolic and Quantitative Approaches with Uncertainty*. Cham: Springer Nature Switzerland, 2023.
3. Nguyen, Vu-Linh, **Haifei Zhang**, and Sébastien Destercke. "Learning Sets of Probabilities Through Ensemble Methods." *European Conference on Symbolic and Quantitative Approaches with Uncertainty*. Cham: Springer Nature Switzerland, 2023.
4. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. "Explaining cautious random forests via counterfactuals." *International Conference on Soft Methods in Probability and Statistics*. Cham: Springer International Publishing, 2022.
5. **Zhang, Haifei**, Benjamin Quost, and Marie-Helène Masson. "Cautious random forests: a new decision strategy and some experiments." *International Symposium on Imprecise Probability: Theories and Applications*. PMLR, 2021.

### National conference papers

1. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. "Explications contrefactuelles pour les forêts aléatoires prudentes." *31ièmes Rencontres Francophones sur la Logique Floue et ses Applications (LFA 2022)*. 2022.

2. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Forêts aléatoires prudentes: une nouvelle stratégie de décision et quelques expériences.” *31ièmes Rencontres Francophones sur la Logique Floue et ses Applications (LFA 2022)*. 2022.

## *Review service*

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|---|---|
| **ECAI** | 2024, 2025 |
| **Forum for Economic and Financial Studies** | 2025 |